

## Section 1: Identification

**MANUFACTURER:** PACE Technologies  
3601 E. 34<sup>th</sup> St.  
Tucson, AZ 85713

**INFORMATION PHONE:** 520-882-6598

**EMERGENCY PHONE:** CHEMTREC 800-424-9300 (US) Day or night  
Customer No. 16568

**TRADE NAME:** Nickel Etchant (Nitric acid, sulfuric acid, NaCl and water)

**CHEMICAL FAMILY:** CORROSIVE LIQUIDS, ACIDIC, INORGANIC, N.O.S.  
(NITRIC ACID, SULFURIC ACID MIXTURE)

**HMIS RATING:** HEALTH: 4 FLAMMABILITY: 3 REACTIVITY: 3

**HAZARD RATING:**

LEAST: 0 SLIGHT: 1 MODERATE: 2 HIGH: 3 EXTREME: 4

## Section 2: Hazard(s) Identification

|                                                                     |                                                                                                                                                                                             |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GHS Classification in accordance with 29 CFR 1910 (OSHA HCS)</b> | Oxidizing liquids (Category 3), H272<br>Corrosive to metals (Category 1), H290<br>Skin corrosion (Category 1A), H314<br>Serious eye damage (Category 1), H318                               |
| <b>PICTOGRAM(s):</b>                                                |                                                                                                          |
| <b>SIGNAL WORD:</b>                                                 | Danger                                                                                                                                                                                      |
| <b>HAZARD STATEMENTS:</b>                                           | <b>Hazard Statement(s):</b><br>H272 - May intensify fire; oxidizer<br>H290- May be corrosive to metals<br>H314- Causes severe skin burns and eye damage<br>H318 - Causes serious eye damage |
| <b>PRECAUTIONARY</b>                                                | <b>Precautionary Statement(s):</b>                                                                                                                                                          |

**STATEMENTS:**

**Preventions:**

P210- Keep away from heat/sparks/open flames/hot surfaces. — No smoking.  
P220- Keep/Store away from clothing and combustible materials.  
P221- Take any precaution to avoid mixing with combustibles, strong bases, metallic powders, carbides, hydrogen sulfide, turpentine, and combustible organics.  
P234- Keep only in original container.  
P280- Wear protective gloves/protective clothing/eye protection/face protection.

**Response:**

P303+P361+P353- IF ON SKIN (or hair): Remove/Take off Immediately all contaminated clothing. Rinse SKIN with water/shower.  
P304+P340- IF INHALED: Remove victim to fresh air and Keep at rest in a position comfortable for breathing.  
P305+P351+P338- IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.  
P310- IF INHALED: Remove victim to fresh air and Keep at rest in a position comfortable for breathing.  
P321- Specific treatment (see Section 4 SDS).  
P363- Wash contaminated clothing before reuse.  
P370+P378- In case of fire: Use dry sand, dry chemical or alcohol-resistant foam for extinction.  
P390- Absorb spillage to prevent material damage.

**Storage:**

P404- Absorb spillage to prevent material damage.  
P405- Store locked up.

**Disposal:**

P501- Dispose of contents/container to Federal, State and Local Regulations.

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**POISON! DANGER! STRONG OXIDIZER. CONTACT WITH OTHER MATERIAL MAY CAUSE FIRE. CORROSIVE. LIQUID AND MIST CAUSE SEVERE BURNS TO ALL BODY TISSUE. MAY BE FATAL IF SWALLOWED OR INHALED. INHALATION MAY CAUSE LUNG AND TOOTH DAMAGE.**

**SAF-T-DATA<sup>(tm)</sup>** Ratings (Provided here for your convenience)

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Health Rating: 4 - Extreme (Poison)  
Flammability Rating: 0 - None  
Reactivity Rating: 3 - Severe (Oxidizer)  
Contact Rating: 4 - Extreme (Corrosive)  
Lab Protective Equip: GOGGLES & SHIELD; LAB COAT & APRON; VENT HOOD; PROPER GLOVES  
Storage Color Code: White (Corrosive)  
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**Potential Health Effects**

**Inhalation:**

Corrosive! Inhalation of vapors can cause breathing difficulties and lead to pneumonia and pulmonary edema, which may be fatal. Other symptoms may include coughing, choking, and irritation of the nose, throat, and respiratory tract.

**Ingestion:**

Corrosive! Swallowing nitric acid can cause immediate pain and burns of the mouth, throat, esophagus and gastrointestinal tract.

**Skin Contact:**

Corrosive! Can cause redness, pain, and severe skin burns. Concentrated solutions cause deep ulcers and stain skin a yellow or yellow-brown color.

**Eye Contact:**

Corrosive! Vapors are irritating and may cause damage to the eyes. Contact may cause severe burns and permanent eye damage.

**Chronic Exposure:**

Long-term exposure to concentrated vapors may cause erosion of teeth and lung damage. Long-term exposures seldom occur due to the corrosive properties of the acid.

**Aggravation of Pre-existing Conditions:**

Persons with pre-existing skin disorders, eye disease, or cardiopulmonary diseases may be more susceptible to the effects of this substance.

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### Section 3: Composition/Information on Ingredients

| Ingredient      | CAS No    | Percent | Hazardous |
|-----------------|-----------|---------|-----------|
| Nitric Acid     | 7697-37-2 | 20-40%  | Yes       |
| Sulfuric Acid   | 7664-93-9 | 2-10%   | Yes       |
| Sodium Chloride | 7647-14-5 | 2-10%   | Yes       |
| Water           | 7732-18-5 | 50-70%  | No        |

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### Section 4: First-Aid Measures Emergency Overview

Immediate first aid treatment reduces the health effects of this substance.

**Inhalation:**

Remove to fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Call a physician.

**Ingestion:**

DO NOT INDUCE VOMITING! Give large quantities of water or milk if available. Never give anything by mouth to an unconscious person. Get medical attention immediately.

**Skin Contact:**

In case of contact, immediately flush skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.

**Eye Contact:**

Immediately flush eyes with plenty of water for at least 15 minutes, lifting lower and upper eyelids occasionally. Get medical attention immediately.

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### Section 5: Fire-Fighting Measures

**Fire:**

Not combustible, but substance is a strong oxidizer and its heat of reaction with reducing agents or combustibles may cause ignition. Can react with metals to release flammable hydrogen gas.

**Explosion:**

Reacts explosively with combustible organic or readily oxidizable materials such as: alcohols, turpentine, charcoal, organic refuse, metal powder, hydrogen sulfide, etc. Reacts with most metals to release hydrogen gas which can form explosive mixtures with air.

**Fire Extinguishing Media:**

Water spray may be used to keep fire exposed containers cool. Do not get water inside container.

**Special Information:**

Increases the flammability of combustible, organic and readily oxidizable materials. In the event of a fire, wear full protective clothing and NIOSH-approved self-contained breathing apparatus with full facepiece operated in the pressure demand or other positive pressure mode.

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## Section 6: Accidental Release Measures

Ventilate area of leak or spill. Wear appropriate personal protective equipment as specified in Section 8. Isolate hazard area. Keep unnecessary and unprotected personnel from entering. Contain and recover liquid when possible. Neutralize with alkaline material (soda ash, lime), then absorb with an inert material (e. g., vermiculite, dry sand, earth), and place in a chemical waste container. Do not use combustible materials, such as saw dust. Do not flush to sewer! US Regulations (CERCLA) require reporting spills and releases to soil, water and air in excess of reportable quantities. The toll free number for the US Coast Guard National Response Center is (800) 424-8802.

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## Section 7: Handling and Storage

Store in a cool, dry, ventilated storage area with acid resistant floors and good drainage. Protect from physical damage. Keep out of direct sunlight and away from heat, water, and incompatible materials. Do not wash out container and use it for other purposes. When diluting, the acid should always be added slowly to water and in small amounts. Never use hot water and never add water to the acid. Water added to acid can cause uncontrolled boiling and splashing. Containers of this material may be hazardous when empty since they retain product residues (vapors, liquid); observe all warnings and precautions listed for the product.

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## Section 8: Exposure Controls/ Personal Protection

**Airborne Exposure Limits:**

For Nitric Acid

-OSHA Permissible Exposure Limit (PEL):

2 ppm (TWA), 4 ppm (STEL)

-ACGIH Threshold Limit Value (TLV):

2 ppm (TWA); 4 ppm (STEL)

For Sulfuric Acid:

- OSHA Permissible Exposure Limit (PEL) -

1 mg/m<sup>3</sup> (TWA)

- ACGIH Threshold Limit Value (TLV) -

0.2 mg/m<sup>3</sup>(T) (TWA) for sulfuric acid - A2 Suspected Human Carcinogen for sulfuric acid contained in strong inorganic mists.

**Ventilation System:**

A system of local and/or general exhaust is recommended to keep employee exposures below the Airborne Exposure Limits. Local exhaust ventilation is generally preferred because it can control the emissions of the contaminant at its source, preventing dispersion of it into the general work area. Please refer to the ACGIH document, *Industrial*

*Ventilation, A Manual of Recommended Practices*, most recent edition, for details.

**Personal Respirators (NIOSH Approved):**

If the exposure limit is exceeded, wear a supplied air, full-facepiece respirator, airlined hood, or full-facepiece self-contained breathing apparatus. Nitric acid is an oxidizer and should not come in contact with cartridges and canisters that contain oxidizable materials, such as activated charcoal. Canister-type respirators using sorbents are ineffective.

**Skin Protection:**

Wear impervious protective clothing, including boots, gloves, lab coat, apron or coveralls, as appropriate, to prevent skin contact.

**Eye Protection:**

Use chemical safety goggles and/or a full face shield where splashing is possible. Maintain eye wash fountain and quick-drench facilities in work area.

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## Section 9: Physical and Chemical Properties

**Appearance:**

Colorless to yellowish liquid.

**Odor:**

Suffocating, acrid.

**Solubility:**

Infinitely soluble.

**Specific Gravity:**

1.41

**pH:**

1.0 (0.1M solution)

**% Volatiles by volume @ 21C (70F):**

100 (as water and acid)

**Boiling Point:**

122C (252F)

**Melting Point:**

-42C (-44F)

**Vapor Density (Air=1):**

2-3

**Vapor Pressure (mm Hg):**

48 @ 20C (68F)

**Evaporation Rate (BuAc=1):**

No information found.

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## Section 10: Stability and Reactivity

**Stability:**

Stable under ordinary conditions of use and storage. Containers may burst when heated.

**Hazardous Decomposition Products:**

When heated to decomposition, emits toxic nitrogen oxides fumes and hydrogen nitrate. Will react with water or steam to produce heat and toxic and corrosive fumes.

**Hazardous Polymerization:**

Will not occur.

**Incompatibilities:**

A dangerously powerful oxidizing agent, concentrated nitric acid is incompatible with most substances, especially strong bases, metallic powders, carbides, hydrogen sulfide, turpentine, and combustible organics.

**Conditions to Avoid:**

Light and heat.

## Section 11: Toxicological Information

Nitric acid: Inhalation rat LC50: 244 ppm (NO2)/30M; Investigated as a mutagen, reproductive effector. Oral (human) LDLo: 430 mg/kg.

| -----\Cancer Lists\-----    |                      |             |               |
|-----------------------------|----------------------|-------------|---------------|
| Ingredient                  | ---NTP Carcinogen--- |             | IARC Category |
|                             | Known                | Anticipated |               |
| Nitric Acid (7697-37-2)     | No                   | No          | None          |
| Sulfuric Acid (7664-93-9)   | No                   | No          | None          |
| Sodium Chloride (7647-14-5) | No                   | No          | None          |
| Water (7732-18-5)           | No                   | No          | None          |

## Section 12: Ecological Information

### Environmental Fate:

No information found.

### Environmental Toxicity:

No information found.

## Section 13: Disposal Considerations

Whatever cannot be saved for recovery or recycling should be managed in an appropriate and approved waste facility. Although not a listed RCRA hazardous waste, this material may exhibit one or more characteristics of a hazardous waste and require appropriate analysis to determine specific disposal requirements. Processing, use or contamination of this product may change the waste management options. State and local disposal regulations may differ from federal disposal regulations. Dispose of container and unused contents in accordance with federal, state and local requirements.

## Section 14: Transportation Information

### Domestic (Land, D.O.T.)

**Proper Shipping Name:** CORROSIVE LIQUIDS, ACIDIC, INORGANIC, N.O.S. (NITRIC ACID, SULFURIC ACID, MIXTURE)

**Hazard Class:** 8

**UN/NA:** UN3264

**Packing Group:** II

**Limited Quantity Shipping** <1 L

### International (Water, I.M.O.)

**Proper Shipping Name:** CORROSIVE LIQUIDS, ACIDIC, INORGANIC, N.O.S. (NITRIC ACID, SULFURIC ACID, MIXTURE)

**Hazard Class:** 8  
**UN/NA:** UN3264  
**Packing Group:** II

### Section 15: Regulatory Information

| -----\Chemical Inventory Status - Part 1\----- |      |     |       |           |
|------------------------------------------------|------|-----|-------|-----------|
| Ingredient                                     | TSCA | EC  | Japan | Australia |
| Nitric Acid (7697-37-2)                        | Yes  | Yes | Yes   | Yes       |
| Sulfuric Acid (7664-93-9)                      | Yes  | Yes | Yes   | Yes       |
| Sodium Chloride (7647-14-5)                    | Yes  | Yes | Yes   | Yes       |
| Water (7732-18-5)                              | Yes  | Yes | Yes   | Yes       |

| -----\Chemical Inventory Status - Part 2\----- |       |            |      |       |
|------------------------------------------------|-------|------------|------|-------|
| Ingredient                                     | Korea | --Canada-- |      |       |
|                                                |       | DSL        | NDSL | Phil. |
| Nitric Acid (7697-37-2)                        | Yes   | Yes        | No   | Yes   |
| Sulfuric Acid (7664-93-9)                      | Yes   | Yes        | No   | Yes   |
| Sodium Chloride (7647-14-5)                    | Yes   | Yes        | No   | Yes   |
| Water (7732-18-5)                              | Yes   | Yes        | No   | Yes   |

| -----\Federal, State & International Regulations - Part 1\----- |            |      |                    |                |
|-----------------------------------------------------------------|------------|------|--------------------|----------------|
| Ingredient                                                      | -SARA 302- |      | -----SARA 313----- |                |
|                                                                 | RQ         | TPQ  | List               | Chemical Catg. |
| Nitric Acid (7697-37-2)                                         | 1000       | 1000 | Yes                | No             |
| Sulfuric Acid (7664-93-9)                                       | 1000       | 1000 | Yes                | No             |
| Sodium Chloride (7647-14-5)                                     | No         | No   | No                 | No             |
| Water (7732-18-5)                                               | No         | No   | No                 | No             |

| -----\Federal, State & International Regulations - Part 2\----- |        |        |        |
|-----------------------------------------------------------------|--------|--------|--------|
| Ingredient                                                      | CERCLA | -RCRA- | -TSCA- |
|                                                                 |        | 261.33 | 8 (d)  |
| Nitric Acid (7697-37-2)                                         | 1000   | No     | No     |
| Sulfuric Acid (7664-93-9)                                       | 1000   | No     | No     |
| Sodium Chloride (7647-14-5)                                     | No     | No     | No     |
| Water (7732-18-5)                                               | No     | No     | No     |

Chemical Weapons Convention: No      TSCA 12(b): No      CDTA: No  
SARA 311/312: Acute: Yes      Chronic: Yes      Fire: Yes      Pressure: No  
Reactivity: No      (Mixture / Liquid)

**Australian Hazchem Code:** 2PE

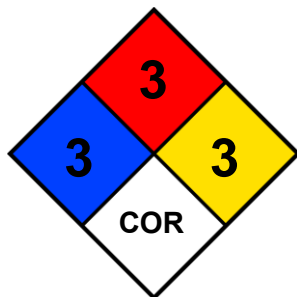
**Poison Schedule:** S6

**WHMIS:**

This MSDS has been prepared according to the hazard criteria of the Controlled Products Regulations (CPR) and the MSDS contains all of the information required by the CPR.

## Section 16: Other Information

### 16.1 NFPA 704



**Top, Flammability: 3 – Severe Hazard**

**Left, Health Hazard: 3 – Severe Hazard**

**Right, Reactivity: 3 – Severe Hazard**

**Bottom, Special Notice: COR – Corrosive**

**Product Use:**

Laboratory Reagent.

**Disclaimer:**

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